Canon Bruce

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EDUCATION

Vanderbilt University Expected Graduation: May 2028
Nashville, TN
B.E. Electrical and Computer Engineering, Minors in Computer Science
Dean's List Spring 2026 | GPA: 3.34/4.00
Relevant Coursework: Electronics, Analog Circuits, Intermediate Software Design, Linear Algebra, Differential Equations

PROFESSIONAL EXPERIENCE

Sitevue AI **Franklin, TN**
Electrical Engineer Intern *June 2026 - August 2026*
- Developed a Linux-based robotics platform integrating computer vision, AI decision-making, and precision arm control to automate tile identification and painting across large-scale industrial environments.
- Collaborated with engineering leadership to develop autonomous robotics solutions and support deployment opportunities.

Legacy Audio **Springfield, IL**
Software Engineer Intern *March 2026 - May 2026*
- Architected a Raspberry Pi-based Qt touchscreen interface with 10+ user controls, replacing physical controls on DSP unit.
- Led end-to-end touchscreen integration efforts, optimizing display configuration, hardware interfacing, and embedded software deployment to deliver a fully functional prototype for live product demonstration.
- Contributed to the transition from outsourced to in-house DSP production, reducing per-unit costs from $2,500 to $1,200.

Vision Care Associates **Springfield, IL**
Lab Technician Intern *May 2024 - August 2025*
- Operated precision optical machinery to fulfill 500+ patient lens prescriptions, maintaining throughput with zero quality escapes.
- Streamlined daily lab workflows by proactively communicating status updates to senior technicians, reducing delays by 20%.
- Cross-trained across multiple stages of the lens fabrication process, increasing operational flexibility within the lab.

Elks Lodge **Rochester, IL**
Head Lifeguard *May 2022 - August 2023*
- Managed schedules and shift coverage for a team of 12 lifeguards, ensuring zero understaffed shifts across full summer season.
- Implemented recurring CPR and first aid training sessions every 2-3 weeks, increasing emergency preparedness.

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

Vanderbilt Quantum Initiative **Nashville, TN**
Research Associate *January 2025 – Present*
- Studying quantum communication principle in satellites as part of a 40-member research initiative at Vanderbilt.
- Contributing to MATLAB-based satellite simulation project exploring quantum communication in low earth orbit systems.
- Synthesized findings from academic literature and industry research to support project direction and technical decision-making.

Theta Tau, Professional Engineering Fraternity, Vanderbilt University **Nashville, TN**
Intramural Chair *August 2025 – Present*
- Coordinate intramural programming across 6 teams and 5 sports for 100+ member professional engineering fraternity.
- Introduced a points-based reward system tying event attendance and team performance to a semester-end prize, incentivizing consistent participation and driving stronger team spirit and engagement across intramural programming.
- Grew chapter intramural participation from 30% to 50% by restructuring team rosters and streamlining scheduling processes.

ECE 2212, Digital Systems, Vanderbilt University **Nashville, TN**
Teaching Assistant *August 2026 - Present*
- Lead weekly lab sections for over 40 students, guiding hands-on hardware debugging and SystemVerilog implementation.
- Run group study sessions each week, helping 20+ students work through digital logic design problems and prepare for exams.

SKILLS & INTERESTS

Technical: Java, C++, Python, Claude Code, JavaScript, HTML, Excel, SPICE, CAD, Verilog, Quartus, MATLAB
Interests: Fitness & human performance, NBA analytics, caffeinated beverage connoisseur, Pokémon, SEC basketball & football